More Than One Type of Number Agreement Computation? Investigating Planning Time-Course through Agreement Attraction in Reflexive Pronoun Production

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1. Introduction English number agreement attraction shows drastically different profiles across dependency types: subject–verb agreement yields many more attraction errors than reflexive–antecedent agreement [1, 2]. If these dependencies are computed by a single mechanism, this asymmetry is surprising. Focusing on **reflexive production**, we ask whether subject–verb and reflexive agreement reflect (i) one retrieval-based mechanism applied at different planning times, or (ii) qualitatively different mechanisms drawing on different sources of number information.

2. Competing Hypotheses The *Planning Order account* posits a single retrieval-based mechanism that accesses number features from already planned linguistic materials [2]. Subject–verb agreement is computed late, after the full subject NP (including the local intervener) has been encoded, whereas reflexive agreement is computed earlier, before the interfering PP is planned [2]. By contrast, the *Number Source account* posits two distinct routes: a "linguistic route" that retrieves number from syntactic representations (used by verbs) and a "conceptual route" that retrieves number directly from the preverbal message (used by reflexives), making reflexives intrinsically robust to attraction [3]. Both accounts capture the classic contrast between subject–verb and coargument reflexives, but they diverge for **non-coargument reflexives**, which has not been investigated.

3. Method We tested four sentence conditions (Table 1): `Verb`, `Coargument`, and two noncoargument reflexives, `Complement` (e.g., *the picture of oneself*) and `Adjunct` (e.g., *the picture next to oneself*). On standard syntactic analyses [5], reflexives in complement PPs are structurally closer to the head noun than adjunct PPs (Figure 3). If planning proceeds over hierarchical constituents, reflexives in `Coargument` and `Complement` positions can be computed before the intervening PP is encoded, whereas `Adjunct` reflexives are computed later, after the full subject NP (including the intervener) is planned. Planning Order thus predicts `Coargument` $\approx$ `Complement` $<$ `Adjunct` and `Verb`. Number Source instead predicts uniformly low attraction for all reflexive conditions, regardless of syntactic position, and a larger contrast between all reflexives and `Verb`.

To elicit naturalistic preverbal message formulation and incremental linguistic material planning, we used a *scene-description paradigm* following [2]. Participants ($N = 88$, after preregistered exclusions) described short picture sequences using a limited lexicon (Figure 1). Each condition has four subject–intervener number pairings (SS, SP, PS, PP; S = singular, P = plural).

4. Results Figure 2 shows error rates by sentence condition. A binomial generalized linear mixed model (Eq. 1) and estimated marginal means replicated the classic contrast between `Coargument` and `Verb`: `Coargument` had the lowest overall error rate, significantly lower than `Adjunct` and `Verb`, with `Complement` in between but not reliably different from `Coargument`. However, *only* `Verb` showed a reliable attraction effect: mismatch (SP/PS) trials yielded significantly more agreement errors than match (SS/PP) trials. All three reflexive conditions showed numerically higher error rates in mismatch than match *but no significant match effect*. All statistics are shown in Table 2.

5. Implications Taken together, these results favor a *Number Source* view over a pure *Planning Order* account. The absence of significant attraction effects for any reflexive condition supports the idea that reflexive agreement primarily retrieves number from conceptual representations, largely insulating it from syntactic attraction. At the same time, the higher baseline error rate in `Adjunct` compared to `Coargument` and `Complement` is a novel finding and suggests that this conceptual route is nonetheless sensitive to syntactic structure, in particular to whether the reflexive appears inside the lowest N′ (head + complement PP) or in a higher adjunct PP (Figure 3). This points to a hybrid agreement mechanism in which both conceptual and linguistic routes contribute to agreement, yielding reflexive planning that is robust to attraction yet modulated by structural factors.

Condition	Sample Sentence
`Coargument`	The blueys above the greeny mimmed {themselves, * itself}.
`Complement`	The blueys above the greeny mimmed the pictures of {themselves, * itself}.
`Adjunct`	The blueys above the greeny mimmed the pictures next to {themselves, * itself}.
`Verb`	The blueys above the greeny {were, * was} mimming.

Table 1: Sample sentences of the critical conditions in plural-singular.

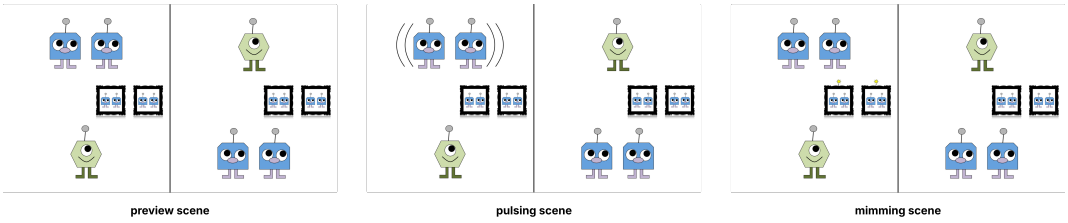

Figure 1: Sample stimuli for *The blueys above the greeny mimmed the pictures of themselves*.

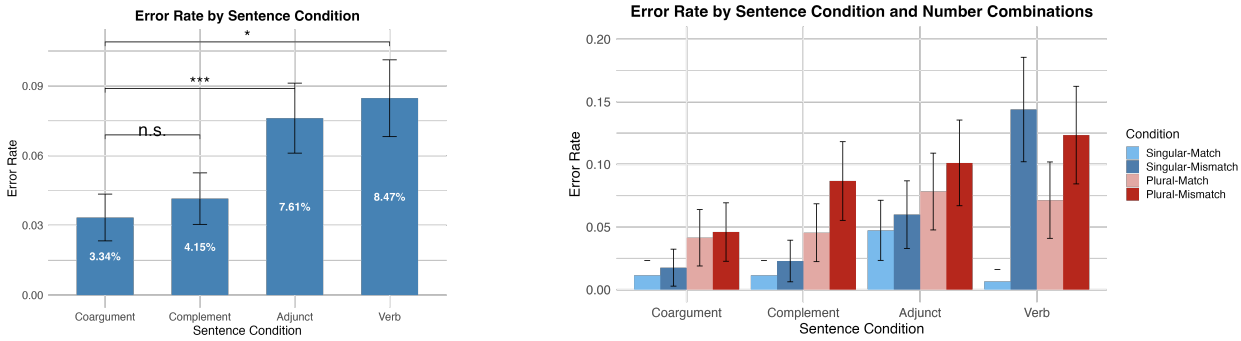

Figure 2: Error rates of the four conditions (left) and subdivided by number features (right).

$$\text{error} \sim \text{match_condition} * \text{antecedent} * \text{sentence_condition} + (1 + \text{match_condition} \mid \text{participant}) \quad (1)$$

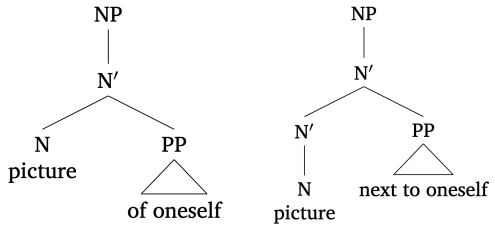

Figure 3: Theoretical analyses of `Complement` (left) and `Adjunct` (right).

Condition	Overall error rate (total)	β	SE	p	EMM est. prob match vs. mismatch
Coargument	2.90% (689)	—	—	—	0.65% vs. 1.37%
Complement	4.15% (699)	0.299	0.377	0.43	0.69% vs. 2.32%
Adjunct	7.16% (670)	1.229	0.326	$< .001$	2.22% vs. 4.35%
Verb	8.63% (614)	1.053	0.402	$< .01$	0.85% vs. 8.71%

Table 2: Overall error rates and GLMM & EMM statistics by condition.

Selected References [1] Dillon et al. (2013). Contrasting intrusion profiles for agreement and anaphora: Experimental and modeling evidence. *JML*. **[2]** Kandel & Phillips (2022). Number attraction in verb and anaphor production. *JML*. **[3]** Kandel et al. (2024). Number attraction in pronoun production. *Open Mind*. **[4]** Eberhard et al. (2005). Making syntax of sense: number agreement in sentence production. *Psychological review*. **[5]** Bryant & Charnavel (2020). Effects of creator role on binding into picture noun phrases. In *Proceedings of the 55th annual meeting of the CLS*. **[6]** Eberhard (1997). The marked effect of number on subject–verb agreement. *JML*.